

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIC602 - ADVANCED MACHINE LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 In K-Means clustering, what does 'K' represent?
 - a) The number of features in the dataset
 - b) The number of outliers detected
 - c) The number of clusters to be formed
 - d) The number of iterations allowed
- 2 The Silhouette Score ranges between:
 - a) -1 to 0
 - b) -1 to 1
 - c) 0 to 1
 - d) 0 to ∞
- 3 In Association Rule Mining, which parameter measures the likelihood that item B is purchased when item A is purchased?
 - a) Lift
 - b) Confidence
 - c) Support
 - d) Leverage
- 4 In Collaborative Filtering, recommendations are based on:
 - a) Product attributes
 - b) User behavior similarity
 - c) Random selection
 - d) Advertisements
- 5 Which of the following represents the Exploration vs Exploitation Dilemma?
 - a) Choosing random actions only
 - b) Avoiding risky actions altogether
 - c) Always selecting the highest reward action
 - d) Balancing between trying new actions and using known good actions
- 6 In Q-Learning, the Q-value represents:
 - a) The immediate reward only
 - b) The expected cumulative future reward for an action in a given state
 - c) The probability of state transitions
 - d) The agent's learning rate
- 7 The parameter α (alpha) in Q-Learning controls:
 - a) Discounting future rewards
 - b) Randomness of action selection
 - c) The learning rate, or how quickly the agent updates its knowledge
 - d) The total number of actions
- 8 A Time Series is said to be stationary when:
 - a) It shows an increasing trend over time
 - b) Its statistical properties like mean and variance remain constant over time
 - c) It has a seasonal pattern
 - d) It is unpredictable

- 9 In an ARMA model, the AR and MA terms represent:
- Auto-Regressive and Moving Average components
 - Automatic Residual and Mean Adjustment
 - Averaged Random Mean Algorithm
 - Absolute Regression and Model Accuracy
- 10 The purpose of Model Selection is to:
- Always choose the simplest model regardless of accuracy
 - Find the model that only fits the training data
 - Select the model that generalizes well to unseen data
 - Select the model that has the most parameters
- 11 Which of the following is NOT a type of Boosting algorithm?
- AdaBoost
 - Gradient Boosting
 - XGBoost
 - K-Means Boosting
- 12 Which of the following is a key reason for using Cross-Validation in model selection?
- To increase overfitting
 - To evaluate model performance on unseen data
 - To reduce data preprocessing steps
 - To perform feature scaling
- Q2. Solve the following.
- Describe the Hierarchical Clustering Approach. Compare Top-down (Divisive) and Bottom-up (Agglomerative) clustering with suitable examples.
 - Explain the intuition behind Principal Component Analysis (PCA). How does PCA help in Dimensionality Reduction? Include a brief explanation of the mathematical steps involved.
- Q3. Solve the following.
- Differentiate between Collaborative Filtering and Content-Based Filtering with appropriate examples. List their advantages and limitations.
 - Explain Association Rules with suitable real-world examples. What are Support, Confidence, and Lift in this context?
- Q4. Solve Any Two of the following.
- Consider an agent using Q-Learning with:
Learning Rate (α) = 0.5, Discount Factor (γ) = 0.9, Current $Q(s, a) = 5$, Immediate Reward (R) = 10, and Maximum Q -value for next state = 7.
Compute the updated $Q(s, a)$ using the Q-Learning formula: $Q(s, a) = Q(s, a) + \alpha [R + \gamma * \max_{a'} Q(s', a') - Q(s, a)]$
 - Explain the Exploration vs Exploitation Dilemma in Reinforcement Learning. How does the Epsilon Greedy Algorithm help to address this dilemma?
 - Describe the components of a Markov Decision Process (MDP). Explain why MDPs are suitable for Reinforcement Learning problems.
- Q5. Solve Any Two of the following.
- Explain the components of Time Series Analysis. Illustrate each component with 6 examples.
 - Differentiate between $AR(p)$ and $MA(q)$ models. Provide mathematical expressions and explain how they are used in Time Series modeling.

- C) Briefly explain the components of an ARIMA(p, d, q) model. How does differencing help in achieving stationarity? Provide a step-by-step outline of ARIMA modeling for a non-stationary series.

Q6. Solve Any Two of the following.

- A) Analyze why Boosting algorithms can convert weak learners into strong learners. Include the role of weighted datasets in your explanation. 6
- B) In the first round of AdaBoost, a weak classifier has the following performance: 6
Total instances: 100 Misclassified instances: 20
Calculate the error rate and alpha (α) for the weak learner
- C) List and explain the different types of Boosting Algorithms. How do they differ in terms of approach and efficiency? 6

*** End ***